U.G.C. NET Ex
COMPUTE
(Solved with Expla

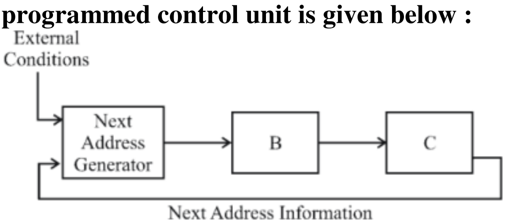
1. Which of the following is an interrupt according to temporal relationship with system clock?
(a) Maskable interrupt
(b) Periodic interrupt
(c) Division by zero
(d) Synchronous interrupt

Ans : (d) Synchronous Interrupt : The source of interrupt is in phase to the system clock in called synchronous interrupt. In other words interrupts which are dependent on the system clock.
Example : Time service that uses the system clock.

2. Which of the following is incorrect for virtual memory?
(a) Large programs can be written
(b) More I/O is required
(c) More addressable memory available
(d) Faster and easy swapping of process

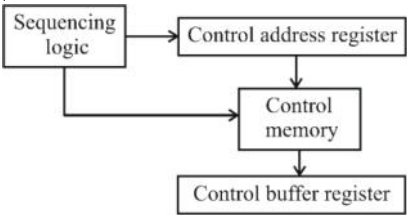
Ans : (b) Virtual Memory : Virtual memory is a memory management technique implemented using both software and hardware. It maps memory address used by program called virtual addresses into physical addresses in computer memory.
It provides more addressable memory, program of larger size than physical memory can be run on same computer but I/O cost is not increases.

3. The general configuration of the micro programmed control unit is given below :



What are blocks B and C in the diagram respectively?
(a) Block address register and cache memory
(b) Control address register and control memory
(c) Branch register and cache memory
(d) Control address register and random access memory

Ans : (b)



So Block B is control address register and block C is control memory.

4. Match the following :

Addressing mode		Location of operand	
A.	**Implied**	**i.**	**Registers which are in CPU**
B.	**Immediate**	**ii.**	**Register specifies the address of the operand**
C.	**Register**	**iii.**	**Specified in the register**
D.	**Register Indirect**	**iv.**	**Specified implicitly in the definition of instruction**

Codes :

	A	B	C	D
(a)	iv	iii	i	ii
(b)	iv	i	iii	ii
(c)	iv	ii	i	iii
(d)	iv	iii	ii	i

Ans : (b)

(i) Implied addressing mode: Specified implicitly in the definition of instruction.

(ii) Immediate addressing mode: Specified in the register.

(iii) Register addressing mode: Registers which are in CPU.

(iv) Register indirect addressing mode: Register specifies the address of the operand.

5. In 8085 microprocessor, the digit 5 indicates that the microprocessor needs

(a) –5 volts, +5 volts supply
(b) +5 volts supply only
(c) –5 volts supply only
(d) 5 MHz clock

Ans : (b) The "5" in the part number highlighted the fact that the 8085 uses a single +5 volt (V). Power supply by using depletion mode transistors, rather than requiring the +5v, –5v and +12v supplies needed by the 8085.

It uses a single +5 volt d.c. (direct current) supply for its operation. It clock speeds is 3 MHz.

6. In 8085, which of the following performs : load register pair immediate operation?

(a) LDAS rp (b) LHLD addr
(c) LXI rp, data (d) INS rp

Ans : (c) LXI is load register pair immediate which take one register as input and 2 bytes immediate data. There are seven integral 8-bit register.
This is generally used to load SP, memory pointer. Generally H–L, D–E, B–C register pairs are used. B, D, H hold the MSB and L, E, C hold the LSB

7. Consider following schedules involving two transactions :
$S_1 : r_1 (X); r_1 (Y); r_2(X);r_2(Y); w_2(Y); w_1(X)$
$S_2 : r_1(X); r_2(X); r_2(Y); w_2(Y); r_1(Y); w_1(X)$
Which of the following statement is true?
(a) Both S_1 and S_2 are conflict serializable.
(b) S_1 is conflict serializable and S_2 is not conflict serilizable
(c) S_1 is not conflict serializable and S_2 is conflict serializable
(d) Both S_1 and S_2 are not conflict serializable

Ans : (c) S_1 : here we can see that two conflicting operations $r_1(Y) \rightarrow w_2(Y)$ & $r_2(X) \rightarrow w_1(X)$, so there will be edge from T_1 to T_2 & T_2 to T_1 .... so there will be loop in the graph, so it is not conflict serializable.
S_2 : But here we are not getting such loop in graph for any edge...... so it is conflict serializable.

8. Which one is correct w.r.t. RDMBS?
(a) primary key $\subseteq$ super key $\subseteq$ candidate key
(b) primary key $\subseteq$ candidate key $\subseteq$ super key
(c) super key $\subseteq$ candidate key $\subseteq$ primary key
(d) super key $\subseteq$ primary key $\subseteq$ candidate key

Ans: (b) Super Key : An attribute that uniquely defines a tuple within a relation.
Candidate key : Super key such that no proper subset is a super key within in a relation so basically has two properties : Each candidate key uniquely identifies tuple in the relation & no proper subset of the composite key has the uniqueness property
Primary Key : Candidate key chosen to identify tuples uniquely within the relation.

9. Let pk(R) denote primary key of relation. R. A many-to-one relationship that exists between two relation R1 and R2 can be expressed as follows :
(a) $pk(R_2) \rightarrow pk(R_1)$
(b) $pk(R_1) \rightarrow pk(R_2)$
(c) $pk(R_2) \rightarrow R_1 \cap R_2$
(d) $pk(R_1) \rightarrow R_1 \cap R_2$

Ans : (b) Many to one relationship set exists between entity sets student & course (a relation & with primary key roll-no) let it is decomposed in 2 relation r1 & r2
So the functional dependency pk(student) $\rightarrow$pk(course) indicates a many to one relationship b/w r1 & r2

10. For a database relation R(A, B, C, D) where the domains of A, B and D include only atomic values, only the following functional dependencies and those that can be inferred from them are :
$A \rightarrow C$
$B \rightarrow D$

The relation R is in ....

(a) First normal form but not in second normal form
(b) Both in first normal form as well in second normal form
(c) Second normal form but not in third normal form
(d) Both in second normal form as well in third normal form

Ans : (a) Here $A \rightarrow C$ & $B \rightarrow D$ no. Singleton attribute derives all other attribute. But if we take $AC \rightarrow BD$ so closure of $AC = \{AC\}^+ = \{A, B, C, D\}$
So AB is the only candidate key of the relation.
Now as 2NF says, attributes should be fully dependent on the key. But here C & D are partially dependent.
So it is in 1NF

11. Consider the following relation :

Works (emp_name, company_name, salary)
Here, emp_name is primary key.
Consider the following SQL query

```
Select emp_name
From works T
where salary > (select avg (salary)
                from works S
               where T.company_name =
                     S.company_name)
```

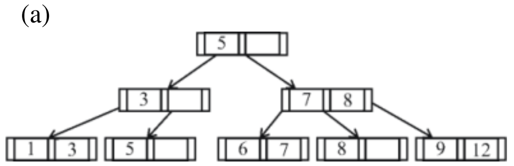
The above query if for following :

(a) Find the highest paid employee who earns more than the average salary of all employees of his company
(b) Find the highest paid employee who earns more than the average salary of all the employees of all companies
(c) Find all employees who earn more than the average salary of all employees of all the companies
(d) Find all employees who earn more than the average salary of all employees of their company

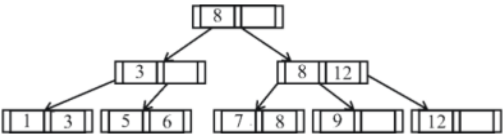
Ans : (d) It doesn't select the highest paid employees, as max is not used, It just select employees name who earns more than the average salary of all employee of the all companies will not be considered.

12. If following sequence of keys are inserted in a B+tree with K(=3) pointers :
8, 5, 1, 7, 3, 12, 9, 6
Which of the following shall be correct B+tree?

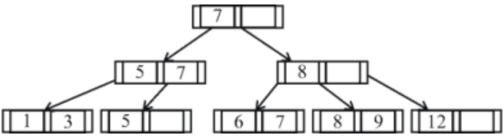
(a)



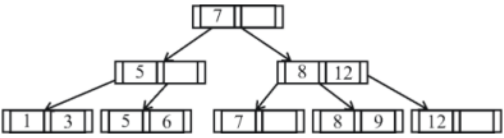
(b)



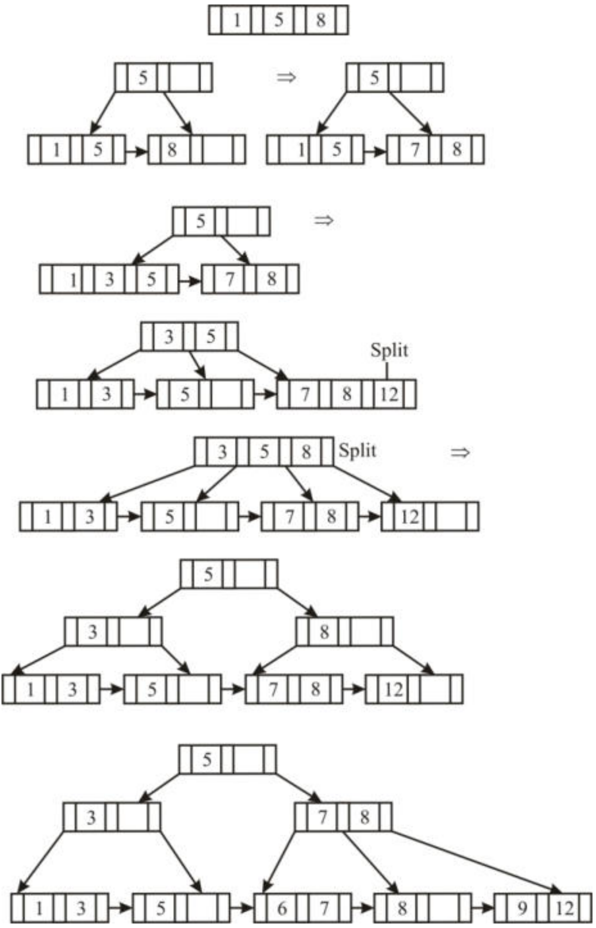
(c)



(d)



Ans : (a) B^+ tree with maximum 3 pointers:



13. Which of the following statement(s) is/are correct?

(a) Persistence is the tern used to describe the duration of phosphorescence

(b) The control electrode is used to turn the electron beam on and off

(c) The electron gun creates a source of electrons which are focused into a narrow beam directed at the face of CRT
(d) All of the above

Ans : (d) Persistence is the time to which the phosphorus in the cathode ray tubes emit light until the next electrons is fired.
Control electrode – an electrode to which a varying signal is applied to vary the output of a transistor. An electron gun is an electrical component is some vacuum tubes that produces a narrow, collimated electron beam that has precise kinetic energy.

14. A segment is any object described by GKS commands and data that start with CREATE SEGMENT and Terminates with CLOSE SEGMENT command. What function can be performed on these segments?
(a) Translation and Rotation
(b) Panning and Zooming
(c) Scaling and Shearing
(d) Translation, Rotation, Panning and Zooming

Ans : (d) A segment is any object described by GKS command and data that start with CREATE SEGMENT and terminates with close segment command.
Segments allows object to be manipulated by simply referring to the segment numbers as a result segments can be translated, rotated, panned & zoomed.

15. Match the following :

A.	**Glass**	**i.**	**Contains liquid crystal and serves as a bonding surface for a conductive coating.**
B.	**Conductive coating**	**ii.**	**Acts as a conductor so that a voltage can be applied across the liquid crystal**
C.	**Liquid crystal**	**iii.**	**A substance which will polarize light when a voltage is applied to it.**
D.	**Polarized film**	**iv.**	**A transparent sheet that polarizes light**

Codes :

	A	B	C	D
(a)	i	ii	iii	iv
(b)	i	iii	ii	iv
(c)	iv	iii	ii	i
(d)	iv	ii	i	iii

Ans : (a) Glass or plastic plate which contains liquid crystal end serves as a bonding surface for conducting coating.

Conductive coating acts as a conductor so that a voltage can be applied across the liquid crystal.

Liquid crystal is a substance which will polarize light when a voltage is applied to it.

Polarized film is a transparent sheet that polarizes light. The axis of polarization is at 90° out of phase with the axis of polarization of liquid crystal.

16. Below are the few steps given for scan-converting a circle using Bresenham's Algorithm.

Which of the given steps is not correct?

(a) Compute d = 3 – 2r (where r is radius)

(b) Stop if x > y

(c) If d < 0, then d = 4x + 6 and x = x + 1

(d) If d ≥ 0, then d = 4 * (x –y) + 10, x = x + 1 and y = y + 1

Ans : (d) Step 1 : Get the coordinates of center of the circle & radius, & store them in X, Y & R respectively, so P = O & Q = R

Step 2 : Set decision parameter D = 3 – 2R

Step 3 : Repeat through set–8 while x<y

Step 4 : Call draw circle (x, y, P, Q)

Step 5 : Increment the value of P

Step 6 : If D < O the D = D + 4x + 6

Step 7 : Else set y = y + 1, D = d + y (x – y) + 10

Step 8 : Call draw circle (x, y, P, Q)

17. Which of the following is/are side effects of scan conversion?

A. Aliasing

B. Unequal intensity of diagonal lines

C. Over striking in photographic applications

D. Local Global aliasing

(a) A and B (b) A, B and C

(c) A, C and D (d) A, B, C and D

Ans : (d)

(i) During scan conversion same pixel may be unnecessarily plotted multiple times which cause **overstriking.**

(ii) Aliasing effect occur due to bad sampling of scan lines over the object, which is occur due to discontinuity of objects and scanning is referred at integer value.

(iii) Unequal intensity occurs due to uneven brightness of object during scan conversion.

(iv) Local and Global aliasing also the side effect of scan conversion.

18. Consider a line AB with A = (0, 0) and B = (8, 4). Apply a simple DDA algorithm and compute the first four plots on this line.

(a) [(0, 0), (1, 1), (2, 1), (3, 2)]

(b) [(0, 0), (1, 1.5), , 2), (3, 2)]

(c) [(0, 0), (1, 1), (2, 2.5), (3, 3)]

(d) [(0, 0), (1, 2), (2, 2), (3, 2)]

Ans : (a) According to **DDA algorithm:**
$|dx| = 8$, $|dy| = 4$
Since $|dx| > |dy|$; so step $= |dx| = 8$

$$X_{inc} = \frac{|dx|}{steps} = \frac{8}{8} = 1$$

$$Y_{inc} = \frac{|dy|}{steps} = \frac{4}{8} = 0.5$$

For K = 1 to steps:

$X_{K+1} = Y_K + X_{inc}$;
$Y_{K+1} = Y_K + Y_{inc}$;

Since DDA algorithm works only for integer values only, we need to round these values of co-ordinates to nearest integer.
So our first four points are (0, 0), (1, 1), (2, 1) and (3, 2).

19. Which of the following are not regular?

(A) Strings of even number of a's
(B) Strings of a's, whose length is a prime number
(C) Set of all palindromes made up of a's and b's
(D) Strings of a's whose length is a perfect square

(a) (A) and (B) only
(b) (A), (B) and (C) only
(c) (B), (C) and (D) only
(d) (B) and (D) only

Ans : (c)(1) Strings of even number of a's is Regular because we can draw finite acceptor (fA) for this.
(2) Strings of a's whose length is a prime number- There are infinite No.
which are prime and we cannot design FA for Infinte language. This is not Regular
(3) Set of all palindromes made up of a's and b's. This is not Regular because we cannot design FA for infinite language.
(4) Strings of a's whose length is a perfect square. This is not Regular because we cannot design FA for infinite language.

20. Consider the language $L_1 = \phi$ and $L_2 = \{1\}$. Which one of the following represents $L_1^* \cup L_2^* L_1^*$?

(a) $\{\in\}$ (b) $\{\in, 1\}$
(c) ϕ (d) 1^*

Ans : (d) $L_1{}^* = \phi^* = \in$
$L_2^* L_1^* = \{1\}. \in = 1^*$
$L_1^* \cup L_2^* L_1{}^* = \in \cup 1^* = 1^*$

21. Given the following statements :

(A) A class of language that is closed under union and complementation has to be closed under intersection
(B) A class of language that is closed under union and intersection has to be closed under complementation

Which of the following options is correct?

(a) Both (A) and (B) are false

(b) Both (A) and (B) are true

(c) (A) is true, (B) is false

(d) (A) is false, (B) is true

Ans : (c) Here we can derive this statement with the help of closure properties

S1 : True as it is the case for regular grammar

S2 : False as in RE language it is closed under union, Intersection but not in complementation.

22. Let G = (V, T, S, P) be a context-free grammar such that every one of its productions if of the form $A \rightarrow v$, with $|v| = K > 1$. The derivation tree for any $W \in L(G)$ has a height h such that

(a) $\log_k |W| \le h \le \log_k \left(\frac{|W|-1}{K-1}\right)$

(b) $\log_k |W| \le h \le \log_k \left(K|W|\right)$

(c) $\log_k |W| \le h \le K \log_k |W|$

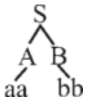
(d) $\log_k |W| \le h \le \left(\frac{|W|-1}{K-1}\right)$

Ans : (d) $S \rightarrow AB$

$A \rightarrow aA \mid aa$

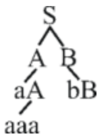
$B \rightarrow bB \mid bb$, Let W = aabb

w = aaabb



$\Rightarrow$ height = 2

$= \log_2^w = \log_2^y = 2$



$\Rightarrow$ height = 3

$3 \le \left(\frac{|w|-1}{k-1}\right)$

$3 \le \frac{5-1}{2-1}$

$3 \le 4$

23. Given the following two languages :

$L_1 = \{a^n b^n \mid n \ge 0, n \ne 100\}$

$L_2 = \{w \in \{a, b, c\}^* \mid n_a(w) = n_b(w) = n_c(w)\}$

Which of the following options is correct?

(a) Both L_1 and L_2 are not context free language

(b) Both L_1 and L_2 are context free language

(c) L_1 is context free language, L_2 is not context free language

(d) L_1 is not context free language, L_2 is context free language

Ans : (c) L1 we can decompose as a^nb^n (n < 100 and $a^nb^n \mid n > 100$)
First part regular, second part DCFL, So it is obviously CFL L2 needs two comparison n(a) = n (b) again n(n) = n(c) so we can not make a PDA for this
So it is CSL but not context free

24. A recursive function h, is defined as follows :

h(m) = k, if m = 0
= 1, if m = 1
= 2 h(m – 1) + 4h (m –2), if m ≥ 2

If the value of h(4) is 88 then the value of k is :

(a) 0 (b) 1
(c) 2 (d) –1

Ans : (c) Given, h(m) = k, if m = 0 = 1, if m = 1
= 2h (m –1) + 4h(m–2), if m > 2, the value of h(4) is 88 the value of k is h (4) = 2h(3) + 4h (2) = 2 (2h(2) + 4h(1) + 4h (2)
= 8h(2) + 8h(1) = 8 (2h(1) + 4h(0)) + 8h = 24h (1) + 32 h (0) = 24 + 32k = 88 = k = 2

25. Suppose there are n stations in a slotted LAN. Each station attempts to transmit with a probability P in each time slot. The probability that only one station transmits in a given slot is ....

(a) $nP(1 – P)^{n–1}$ (b) nP
(c) $P(1 – P)^{n–1}$ (d) $n^P(1 – P)^{n–1}$

Ans : (a) It can be derived using binomial theorem of probability
$NC_i\,(p)^i\,(1 – p)^{n–i}$
now 1 stations is sending with probability p, & n–1 stations are not sending with probability 1 – p
now the probability is = $nc_1(p)^1\,(1–p)^{n–1}$
$= np(1 –p)^{n–1}$

26. Station A uses 32 bytes packets to transmit messages to station B using sliding window protocol. The round trip delay between A and B is 40 milliseconds and the bottleneck bandwidth on the path between A and B is 64 kbps. The optimal window size of A is ....

(a) 20 (b) 10
(c) 30 (d) 40

Ans : (b) Bottleneck bandwidth = 64 Kbps = 64×10^3 bps
size of frame = 32×8
Round trip delay = 40 ms = 40×10^{-3} sec
Total data $40 \times 64 \times 10^3 \times 10^{-3}$ = 40 × 64/8 bytes = 320 bytes
1 packet size = 32 byte
no. of packets 320/32 = 10

27. Let G(x) be generator polynomial used for CRC checking. The condition that should be satisfied by G(x) to correct odd numbered error bits, will be :

(a) $(1 + x)$ is factor of $G(x)$
(b) $(1 - x)$ is factor of $G(x)$
(c) $(1 + x^2)$ is factor of $G(x)$
(d) x is factor of $G(x)$

Ans : (a) In this case polynomial generator should satisfy
(A) Polynomial generated shouldn't be divisible by X.
(B) It should be divisible by 1 + x i.e 1 + x is a factor of polynomial

28. In a packet switching network, if the message size is 48 bytes and each packets contains a header of 3 bytes. If 24 packets are required to transmit the message, the packet size is.........

(a) 2 bytes (b) 1 bytes
(c) 4 bytes (d) 5 bytes

Ans : (d) 24 Packets requires to carry 48 byte of data, so each packet carries 48/24 = 2 byte, now each packet has to + 3 byte of header, so packet size = 2 + 3 = 5 byte

29. In RSA public key cryptosystem suppose n = p * q where p and q are primes. (e, n) and (d, n) are public and private keys respectively. Let M be integer such that o < M < n and $\phi(n) = (p - 1)(q - 1)$.
Which of the following equations represent RSA public key cryptosystem?

I. $C \equiv M^e \pmod n$
$M \equiv (C)^d \pmod n$
II. $ed \equiv 1 \pmod n$
III. $ed \equiv 1 (\text{mod } \phi(n))$
IV. $C \equiv M^e (\text{mod } \phi(n))$
$M \equiv C^d (\text{mod } \phi(n))$

Codes :
(a) I and II (b) I and III
(c) II and III (d) I and IV

Ans : (b) I is true because
Encrypted -Text = $(\text{Plain - Text})^e$ mod n
Plain - Text = $(\text{Encrypted - Text})^d$ mod n
III is True because
$d^{-1} = e \bmod \phi(n)$
or $ed = 1 \bmod \phi(n)$
II and IV is wrong.

30. A node X on a 10 Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2 Mbps. Token bucket is initially filled with 16 megabits. The maximum duration taken by X to transmit at full rate of 10 Mbps is .... secs.

(a) 1 (b) 2
(c) 3 (d) 4

Ans : (b) Using T = C ans comes out to be 2 sec (where C Rout – Rin
maximum duration taken by x for Transmission
= b/(m - r)
= 16/ 10 - 2 = 2sec

31. The asymptotic upper bound solution of the recurrence relation given by

$$T(n) = 2T\left(\frac{n}{2}\right) + \frac{n}{\lg n} \text{ is:}$$

(a) O(n2) (b) O(n lg n)
(c) O(n lg lg n) (d) O(lg lg n)

Ans:(c) By master's theorum. A = 2, B = 2, k= 1, P= –1
AS a = bk & p = –1
answer is o(n log2 log logn) = o (n lg log n)

32. Any decision tree that sorts n elements has height ....

(a) Ω(lg n) (b) Ω(n)
(c) Ω(n lg n) (d) Ω (n^2)

Ans : (c) Any decision tree that sorts n elements has height omega (n lg n) proof :
A binary tree of height h has <=2 ∧ h leaves
At least n : leaves are required for sorting
Thus n! < = 2 ∧h
h > = lg (n!) = sum – {i = 2} ∧n lg (i) > sum – {i = n | 2} ∧n lg (n | 2)
>(n | 2) lg (n | 2) = omega (n lg n)

33. Red–black trees are one of may search tree schemes that are "balanced" in order to guarantee that that basic dynamic-set operations take .... time in the worst case.

(a) O(1) (b) O(lg n)
(c) O(n) (d) O(n lg n)

Ans : (b) Red black trees are one of many search tree schemes that are "balance". in order to guarantee that basic dynamic set operations o(lg n) time in worst case.

34. The minimum number of scalar multiplication required, for parenthesization of a matrix-chain product whose sequence of dimension for four matrices is <5, 10, 3, 12, 5> is

(a) 630 (b) 580
(c) 480 (d) 405

Ans : (d) ((5 × 10, 10 × 3). (3 × 12. 12 × 5))
5 × 10 × 3 + 3 × 12 × 5 + 5 × 3 × 5 = 405

35. Dijkstra's algorithm is based on

(a) Divide and conquer paradigm
(b) Dynamic programming
(c) Greedy Approach
(d) Backtracking paradigm

Ans : (c) Dijkstra always choose the closest vertex in V–S to add to set S where V = vertex set & S = set of vertices whose final path shortest path weights from the source have been obtained.
So Dijakstra algorithm is a greedy algorithm.

36. Match the following with respect to algorithm paradigms :

	List-I		List-II
A.	**Merge sort**	**i.**	**Dynamic programming**

B.	**Huffman coding**	**ii.**	**Greedy approach**
C.	**Optimal polygon triangulation**	**iii.**	**Divide and conquer**
C.	**Subset sum problem**	**iv.**	**Back tracking**

Codes :

	A	B	C	D
(a)	iii	i	ii	iv
(b)	ii	i	iv	iii
(c)	ii	i	iii	iv
(d)	iii	ii	i	iv

Ans : (d) Merge sort based on divide and conquer paradigm.

Huffman coding based on greedy approach.

Subset sum problem based on back-tracking.

Optimal polygon triangulation base on dynamic programming.

37. Abstraction and encapsulation are fundamental principles that underlie the object oriented approach to software development. What can you say about the following two statements?

I. Abstraction allows us to focus on what something does without considering the complexities of how it works.

II. Encapsulation allows us to consider complex ideas while ignoring irrelevant detail that would confuse us.

(a) Neither I nor II is correct

(b) Both I and II are correct

(c) Only II is correct

(d) Only I is correct

Ans : (a) Abstraction & Encapsulation are fundamental principles that underlie the object oriented approach to s/w development. Encapsulation allows us to focus on what something does without considering the complexities of how it works.

Abstraction allows us to consider complex ideas while ignoring irrelevant details would confuse us.

Here none is correct.

38. Given the array of integers 'array' shown below :

13	7	27	2	18	33	9	11	22	8

What is the output of the following JAVA statements?

```
int [ ] p = new int [10];
int [ ] q = new int [10];
for (ink k = 0; k < 10; k++)
        p[k] = array [k];
```

q = p;

p[4] = 20;

System out. println (array [4]+ "." + q[4]);

(a) 20 : 20 (b) 18 : 18

(c) 18 : 20 (d) 20 : 18

Ans : (c) For (int k = 0; k < 10; k ++)
p[k] = array [k]; in this loop we are copying all the values from the array to the array P.
Now, Q = P, Q is pointing P, nom, P[4] = 20 but Q contains the old P array where in 4th index i.e. Q[4] is also Q0.
Note : here pointer is involved, so the location is being update, so Q is getting 20 here but the question is about array [4] which was initial array, so value 18
so answer is 18 : 20

39. Consider the following JAVA program :

```
public class First {
    public static int CBSE (int x) {
        if (x < 100) x = CBSE (x + 10)
        return (x – 1)
    }
  public static void main (String [ ] args) {
        system. out. print (First. CBSE (60));
  }
```

What does this program print ?

(a) 59 (b) 95

(c) 69 (d) 99

Ans : (b)

CBSE (60) 95

CBSE (70) 96

CBSE (80) 97

CBSE (90) 98

CBSE (100) 99

Answer is option (b)

40. Which of the following statements(s) with regard to an abstract class in JAVA is/are TRUE?

I. An abstract class is one that is not used to create objects.

II. An abstract class is designed only to act as a base class to be inherited by other classes.

(a) Only I (b) Only II

(c) Neither I nor II (d) Both I and II

Ans : (d) A class that is declared as abstract is known as abstract class. It needs to be extended and its method implemented. It cannot be instantiated.
An abstract class is designed only to act as a base class to be inherited by other class. It is a design concept in program development & provides a base upon which other classes can be built.

41. Which of the following HTML code will affect the vertical alignment of the table content?

(a) <td style = "vertical-align : middle" > Text Here </td>
(b) <td valign = "centre"> Text Here </td>
(c) <td style = "text-align : center"> Text Here </td>
(d) <td align = "middle" > Text Here </td>

Ans : (b) Vertical alignment of the table contents:
<td valign = 'top |middle | bottom | baseline'>
Text here </td>
So, no option matching if middle = Centre, then option (b) is matching.

42. What can you say about the following statements?

I. XML tags are case-insensitive
II. In Java Script, identifier names are case-sensitive.
III. Cascading Style Sheets (CSS) cannot be used with XML.
IV. All well-formed XML documents must contain a document type definition.

(a) Only I and II are false
(b) Only III and IV are false
(c) Only I and III are false
(d) Only II and IV are false

Ans : (d) 1 is false : XML tags are case sensitive
2 : is true : All Java scripts identifiers are case sensitive
3 : is false : Cascading style sheets (CSS) is a style sheet language & can be applied to any XML document
4 : is false : well formed XML document does not need a DTD.

43. Which of the following statements(s) is/are TRUE with regard to software testing?

I. Regression testing technique ensures that the software product runs correctly after the changes during maintenance.
II. Equivalence partitioning is a white-box testing technique that divides the input domain of a program into classes of data from which test cases can be derived.

(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II

Ans : (a) Regression testing is a type of s/w testing that verifies that s/w previously developed & tested still performs correctly even after it was changed with other software.
Equivalence portioning : is Black box testing method to divide input domain into categories for better test coverage.

44. Which of the following are facts about a top-down software testing approach?

I. Top-down testing typically requires the tester to build method stubs.

II. **Top-down testing typically requires the tester to build test drivers.**

(a) Only I (b) Only II

(c) Both I and II (d) Neither I nor II

Ans : (a) Top Down Testing : In this approach testing is conducted from main module to sub module. If the sub module is not developed a temporary program called STUB, used fro simulate the sub module.

Bottom up testing : In this approach testing is conducted from sub module to main module, if the main module is not developed a temporary program called DRIVERS is used to simulate main module.

45. Match the terms related to software Configuration Management (SCM) in List-I with the descriptions in List-II

List-I		List-II	
I.	**Version**	**A.**	**An instance of a system that is distributed to customers**
II.	**Release**	**B.**	**An instance of a system which is functionally identical to other instances, but designed for different hardware/ software configurations**
III.	**Variant**	**C.**	**An instance of a system that differs, in some way, from other instances.**

Codes

	I	II	III
(a)	B	C	A
(b)	C	A	B
(c)	C	B	A
(d)	B	A	C

Ans : (b) Version is an instance of a system that differs, in some way from other instances.

Release is an instance of a system that is distributed to customers.

Variant is an instance of system which is functionally identical to other instances, but designed for different hardware/ software configurations.

46. A software project was estimated at 352 function points (FP). A four person team will be assigned to this project consisting of an architect, two programmers, and a tester. The salary of the architect is ₹ 80,000 per month, the programmer ₹ 60,000 per month and the tester ₹ 50,000 per month. The average productivity for the team is 8 FP per person month. Which of the following represents the projected cost of the project?

(a) ₹ 28,16,000 (b) ₹ 20,90,000

(c) ₹ 26,95,000 (d) ₹ 27,50,000

Ans : (d) Total 352 FP

average 8 FP per person per month

Total 4 person
So $352/(8 \times 4) = 11$
now, (1 architect + 2 programmer + 1 tester)* 11
$= (80000 + 2 \times 60000 + 50000)* 11 = 2750000$

47. Complete each of the following sentences in List-I on the left hand side by filling in the word or phrase from the List-II on the right hand side that best completes the sentence :

List-I		List-II	
I.	**Determining whether you have built the right system is called**	**A.**	**Software testing**
II.	**——— Determining whether you have built the system right is called**	**B.**	**Software verification**
III.	**——— is the process of demonstrating the existence of defects or providing confidence that they do not appear to be present.**	**C.**	**Software debugging**
IV.	**——— is the process of discovering the cause of a defect and fixing it.**	**D.**	**Software validation**

Codes :

	I	II	III	IV
(a)	B	D	A	C
(b)	B	D	C	A
(c)	D	B	C	A
(d)	D	B	A	C

Ans : (d) **1. Validation :** ensures that the whole s/w meets the customer requirements. As we building the right product?
2. Verification : Ensures that part of s/w correctly implements a specific function. As we are building the product right?
3. Testing : Demonstrating the existence of reject or providing confidence that they do not appear to be present.

48. A software company needs to develop a project that is estimated as 1000 function points and planning to use JAVA as the programming language whose approximate lines of b = 1.0 as exponention factor for the basic COCOMO effort equation and c = 3.0 as equation, approximately how long does the project take to complete?

(a) 11.2 months (b) 12.2 months
(c) 13.2 months (d) 10.2 months

Ans : (b) For basic COCOMO model
E effort = a $(KLOC)^b$ in person months
here a = 1.4, b = 1.0 total LOC = 1000 × 50 = 50, 000
= 50 KLOC
So E = 1.4 × 50 = 70
D(Development) time = $C(E)^d$ in months
here C = 3, d = 0.33, so development time = $3 \times (70)^{0.33}$
3 × (approx cube root root of 70) = 3 × 4.1 = 12.3 (approx)

49. A memory management system has 64 pages with 512 bytes page size. Physical memory consists of 32 page frames. Number of bits required in logical and physical address are respectively :
(a) 14 and 15 (b) 14 and 29
(c) 15 and 14 (d) 16 and 32

Ans : (c) We know, number of page = virtual address space/page size
So virtual address space = number of page × page size
$= 2^6 \times 2^9 = 2^{15}$
So we need is bit fo VA & physical address = page size × frame = $2^9 \times 2^5 = 2^{15}$
So 14 bit required for PA

50. Consider disk queue with I/O requests on the following cylinders in their arriving order ;
6, 10, 12, 54, 97, 73, 128, 15, 44, 110, 34, 45
The disk head is assumed to be at cylinder 23 and moving in the direction of decreasing number of cylinders. Total number of cylinders in the disk is 150. The disk head movement using SCAN-scheduling algorithm is :
(a) 172 (b) 173
(c) 227 (d) 228

Ans : (b) SCAN scheduling Algorithm is also known as Elevator algorithm it scans from the one end to other end.
In this case head starts from 23 onwards and then moves to the another end i.e. 150
Here total head movement = 23 + 1 + .1 to 150
= 23 + 150
= 173

51. Match the following for Unix file system :

List-I		List-II	
A.	**Book block**	**i.**	**Information about file system, free block list, free inode list etc.**
B.	**Super block**	**ii.**	**Contains operating system files as well as program and data file created users**

C.	**Inode block**	**iii.**	**Contain boot program and partition table.**
D.	**Data block**	**iv.**	**Contain a table fro every file in the file system Attributes of files are stored here.**

Codes:

	A	B	C	D
(a)	iii	i	ii	iv
(b)	iii	i	iv	ii
(c)	iv	iii	ii	i
(d)	iv	iii	i	ii

Ans: (b) Boot block contains boot program and partition table.

Super block contain information about file system, free block list, free inode list etc.

Inode block contains a table for every file in the file system and attribute of files are stored in it.

Data block contains operating system files as well as program and data files created by users.

52. Some of the criteria for calculation of priority of a process are :

A. Processor utilization by an individual process

B. Weight assigned to a user or group of users

C. Processor utilization by a user or group of processes

In fair share scheduler, priority is calculated based on :

(a) Only (A) and (B) (b) Only (A) and (C)
(c) (A), (B), and (C) (d) Only (B) and (C)

Ans: (c) Fair-share scheduling is a scheduling algorithm for computer operating system is which the CPU usage is equality distributed among system users or groups, as apposed to equal distribution among processes.

53. One of the disadvantages of user level threads compared to Kernel level threads is

(a) If a user level thread of a process executes a system call, all threads in that process are blocked
(b) Scheduling is application dependent
(c) Thread switching doesn't require kernel mode privileges
(d) The library procedures invoked fro thread management is user level threads are local procedures

Ans : (a) User level threads

Advantage

- Thread switching does not require kernel mode privileges
- User level thread can run on any OS.
- User level threads are fast to create & manage.

Disadvantage

- In a typical OS, most system calls are blocking
- Multithreaded application cannot take advantage of multiprocessing.

54. Which statements is not correct about "init" process in Unix?

(a) It is generally the parent of the login shell
(b) It has PID 1.
(c) It is the first process in the system.
(d) Init forks and execs a 'getty' process at every port connected to a terminal.

Ans : (c) Init is the second process first processed is schedule

55. Consider following two rules R1 and R2 in logical reasoning in Artificial Intelligence (AI):

R1 : from $\alpha \supset \beta$

$\dfrac{\textbf{and}\alpha}{\textbf{inter}\beta}$ **is known as Modus Tollens (MT)**

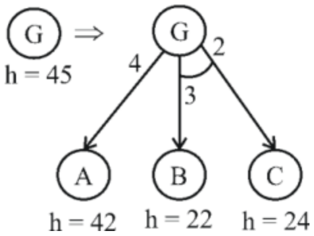
R2 : Form $\alpha \supset \beta$

$\dfrac{\textbf{and}\neg\alpha}{\textbf{inter}\neg\beta}$ **is known as Modus Ponens (MP)**

(a) Only R1 is correct
(b) Only R2 is correct
(c) Both R1 and R2 are correct
(d) Neither R1 nor R2 is correct

Ans: (d) R1 holds the correct definition for modus Ponens & R2 hold the correct definition of modus tollens.

56. Consider the following AO graph :



Which is the best node to expand next by AO* algorithm ?

(a) A (b) B
(c) C (d) B and C

Ans: (a) f(n) = c(n) + h(n)
where,
f(n) = Best path to reach the destination node
c(n) = cost of path
h(n) = heuristic value of a node

cost of choosing B or C is
= (22 + 3) + (24 + 2)
= 51
Note:- We add cost of B and C because they belongs to AND graph
cost of choosing A = 42+ 4 = 46
Since cost of choosing node choosing B or C that why we will expand 'A'.

57. In Artificial Intelligence (AI), what is present in the planning graph?

(a) Sequence of levels (b) Literals
(c) Variables (d) Heuristic estimates

Ans : (a) A planning graph consists of sequence of levels correspond to time steps

- Used to achieve heuristic estimates
  - A solution can also directly be extracted using GRAPHPLAN
- Consist of a sequence of levels that correspond to steps in plan
  - Level 0 is the initial stage

Each level consist of a set of literals and a set of function

- **Literals :** All those that could be true at that time step, depending upon the actions executed at preceding time step.
- **Action :** All those action that could have their preconditions satisfied at that time step, depending on which of the literals actually hold.

58. What is the best method to go for the game playing problem?

(a) Optimal search (b) Random search
(c) Heuristic search (d) Stratified search

Ans : (c) We use a heuristic approach, as it will find out brute force computation, looking at hundreds of thousand of positions.

59. Which of the following statements is true?

(a) The sentence S is a logical consequence of S_1, ...S_n if and only if $S_1 \wedge S_2 \wedge$ .... $\wedge$ $S_n \rightarrow S$ is satisfiable.
(b) The sentence S is a logical consequence of S_1,...........S_n if and only if $S_1 \wedge S_2 \wedge$......... $\wedge$ $S_n \rightarrow S$ is valid.
(c) The sentence S is a logical consequence of S_1,..........S_n if and only if $S_1 \wedge S_2 \wedge$..........$\wedge$ $S_n \wedge \neg S$ is consistent.
(d) The sentence S is a logical consequence of S_1,.........S_n if and only $S_1 \wedge S_2 \wedge$........$\wedge$ $S_n \wedge S$ is inconsistent.

Ans : (b) The sentence S is logical consequence of S_1, S_2, S_3 ….S_n if and only if $S_1 \wedge S_2 \wedge S_3$ …. $\wedge$ $S_n \rightarrow S$ is valid.
Since satisfiable always not implied to valid.

60. The first logic (FOL) statement $(R \vee Q) \wedge (P \vee \neg Q)$ is equivalent to which of the following?

(a) $((R \vee \neg Q) \wedge (P \vee \neg Q) \wedge (R \vee P))$
(b) $((R \vee Q) \wedge (P \vee \neg Q) \wedge (R \vee P))$
(c) $((R \vee Q) \wedge (P \vee \neg Q) \wedge (R \vee \neg P))$
(d) $((R \vee Q) \wedge (P \vee \neg Q) \wedge (\neg R \vee P))$

Ans : (b) P is equivalent to Q means P is true, Q is true & P is false, Q is false
If $(p \vee Q)$ is true and $(P \vee \sim Q)$ is true we can concluded $(P \vee R)$ is also true $(R \vee Q) \Leftrightarrow (\sim R \rightarrow Q)$ similarly
$(P \vee \sim Q) \Leftrightarrow (Q \rightarrow P)$
(1) $(\sim R \rightarrow Q)$, (2) $(Q \rightarrow P)$
$\therefore$ $(\sim R \rightarrow P)$ from 1 & 2 by applying hypothetical syllogism.

$(\sim R \rightarrow P) \Leftrightarrow (R \vee P)$
So $((R \vee Q) \vee (P \vee \neg Q) \vee (R \vee P)((R \vee Q) \wedge (P \vee \neg Q) \wedge (R \vee P)$ is true whenever $((R \vee Q) \wedge (P \vee \neg Q)$ is true also $((R \vee Q) \wedge (P \text{ N } \neg Q))$ $((R \vee Q) \wedge (P \wedge \neg Q) \wedge (R \vee P)$ is false whenever $((R \vee Q) \wedge (P \wedge \neg Q)$ is false.

61. Given the following two statements :
A. L = {w| na(w) = nb(w) is deterministic context free language, but not linear.
B. L = {an bn} is $\cup$ {an b2n} is linear, but not deterministic context free language.
Which of the following options is correct?
(a) Both (A) and (B) are false
(b) Both (A) and (B) are true
(c) (A) is true, (B) is false
(d) (A) is false, (B) is true

Ans : (b) L = {w | na (w) = nb (w)} L = {w | naw} = nb (w)} is deterministic context free language, but not linear
L = { anbn} u} anp 2m} L = {an bn} u {anb2n}
is linear, but not deterministic context free language

62. Which of the following pairs have different expressive power?
(a) Single-tape-turning machine and multi-dimensional turning machine
(b) Multi-tape turning machine and multi-dimensional turning machine
(c) Deterministic push down automata and non-deterministic pushdown automata.
(d) Deterministic finite automata and Non-deterministic finite automata

Ans: (c) Deterministic push down automata & non deterministic push down automata.

63. Which of the following statements is false?
(a) Every context-sensitive language is recursive
(b) The set of all languages that are not recursively enumerable is countable
(c) The family or recursively enumerable language is closed under union.
(d) The families of recursively enumerable and recursive language are closed under reversal

Ans: (b) 1– CSL is a subset of recursive language 3 & 4 are their respective closure properties.

64. Let C be a binary linear code with minimum distance 2t + 1 then it can correct upto........... bits of error.

(a) t + 1 (b) t
(c) t – 2 (d) t/2

Ans : (b) In a set of code words with hamming distance 2++1. t errors can be corrected.

65. A t-error correcting q-nary linear code must satisfy :

$$\mathbf{M\sum_{i=0}^{t}\binom{n}{i}(q-1)^{i} \le X}$$

Where M is the number of code words and X is:

(a) q^n (b) q^t
(c) q^{-n} (d) q^{-t}

Ans : (b) A linear code of length n & rank k is a linear subspace with dimension k of the vector space {\display style\math bb{f} – {Q} ^ {n}} f^n_q where {\display style \math bb {f}–{3} FQ is the finite filed with Q element such a code is called a Q–any code. If Q = 2 or Q = 3, the code is described as a binary code, or a ternary code respectively. The vectors in c are called code words. The size of a code is the number of code words & evals Q^k

66. Names of some of the Operating systems are given below :

(A) MS–DOS
(B) XENIX
(C) OS/2

In the above list, following operating systems didn't provide multi-user facility.

(a) (A) only (b) (A) and (B) only
(c) (B) and (C) only (d) (A), (B) and (C)

Ans : (d)
(i) MS-DOS originally single user, multi-user support provided that did not have present when manufactured.
(ii) XENIX always been multi-user.
(iii) OSI2 does not provide facility of multi-user.
So, none of the O.S. provide facility of multi-user.

67. From the given data below :

a b b a a b b a a b

Which one of the following is not a word in the dictionary created by LZ-coding (the initial words are a, b)?

(a) a b (b) b b
(c) b a (d) b a a b

Ans : (c) Input is : abb aa bb aab
So the dictionary contains : a/b/ba/ab/baa/b

68. With respect to a loop in the transportation table, which one of the following is not correct?

(a) Every loop has an odd no. of cells and atleast 5.
(b) Closed loops may or may not be square in shape
(c) All the cells in the loop that have a plus or minus sign, except the starting cell, must be occupied cells

(d) Every loop has an even no. of cells and atleast four.

Ans : (a) In transportation table:
(i) An even number of atleast 4 cells must participate in closed loop and an occupied cell can be considered only once and not more.
(ii) Closed loop may or may not be square or rectangle in shape.
(iii) All cells in the loop that have a plus or minus sign, except the starting cell, must be occupied.

69. At which of the following stage(s), the degeneracy do no occur in transpiration problem?
(m, n represents number of sources and destinations respectively)
(A) While the values of dual variables u_i and v_j cannot be computed.
(B) While obtaining an initial solution, we may have less than m + n –1 allocations.
(C) At any stage while moving towards optimal solution, when two or more occupied cells with the same minimum allocation become unoccupied simultaneously.
(D) At a stage when the no. of +ve allocation is exactly m + n –1.
(a) (a), (B) and (C) (b) (A), (C) and (D)
(c) (A) and (D) (d) (A), (B), (C) and (D)

Ans: (c) Degeneracy do no occur in transportation problem while the value of dual variables *ui* and *vj* cannot computed and number of positive allocation is exact m + n – 1 (it must be less than m + n – 1).

70. Consider the following LPP :
Min. Z = x1 + x2 + x3
Subject to 3x1 + 4x3 ≤ 5
5x1 + x2 + 6x3 = 7
8x1 + 9x3 ≥ 2,
x1, x2, x3 ≥ 0
The standard form of this LPP shall be :

(a) Min. Z = $x_1+x_2+x_3+0x_4+0x_5$
Subject to $3x_1+4x_3+x_4 = 5;$
$5x_1+x_2+6x_3 = 7;$
$8x_1+9x_3-x_5 = 2;$
$x_1, x_2, x_3, x_4, x_5 \geq 0$

(b) Min. Z = $x_1+x_2+x_3+0x_4+0x_5-1(x_6) - 1x_7)$
Subject to $3x_1+4x_3+x_4 = 5;$
$5x_1+x_2+6x_3 = 7;$
$8x_1+9x_3-x_5 = 2;$
x_1, to $x_7 \geq 0$

(c) Min. Z = $x_1+x_2+x_3+0x_4+0x_5+0x_6$
Subject to $3x_1+4x_3+x_4 = 5;$
$5x_1+x_2+6x_3 = 7;$
$8x_1+9x_3-x_5+x_6 = 2;$
x_1 to $x_6 \geq 0$

(d) Min. Z = $x_1+x_2+x_3+0x_4+0x_5+0x_6+0x_7$
Subject to $3x_1+4x_3+x_4 = 5$;
$5x_1+x_2+6x_3+x_6 = 7$;
$8x_1+9x_3-x_5+x_7 = 2$;
x_1 to $x_7 \geq 0$

Ans : (a) To convert into standard for we need to convert <= and >= equations into = from this either some extra. Variable are added or subtracted from given equation & that variable with O coefficient is included in objective function.
So $3 \times 1\ 4 \times 3 \times 4 = 5$ (Stack variable × 4 is added)
$5 \times 1 + \times 2 + 6 \times 3 = 7$ (already in = from)
$8 \times 1 + 9 \times 3 - \times 3 = 2$ (surplus ×5 is subtracted) now objective function becomes.
minimize $\times 1 + \times 2 + \times 3 + 0 \times 4 + 0 \times 5$

71. Let R and S be two fuzzy relations defined as:

$$\mathbf{R} = \begin{array}{c|cc} & y_1 & y_2 \\ \hline x_1 & 0.6 & 0.4 \\ x_2 & 0.7 & 0.3 \end{array} \text{ and } \mathbf{S} = \begin{array}{c|ccc} & z_1 & z_2 & z_3 \\ \hline y_1 & 0.8 & 0.5 & 0.1 \\ y_2 & 0.0 & 0.6 & 0.4 \end{array}$$

Then, the resulting relation, T, which relates elements of universe x to the elements of universe z using max-min composition is given by :

(a) $T = \begin{array}{c|ccc} & z_1 & z_2 & z_3 \\ \hline x_1 & 0.4 & 0.6 & 0.4 \\ x_2 & 0.7 & 0.7 & 0.7 \end{array}$

(b) $T = \begin{array}{c|ccc} & z_1 & z_2 & z_3 \\ \hline x_1 & 0.4 & 0.6 & 0.4 \\ x_2 & 0.8 & 0.5 & 0.4 \end{array}$

(c) $T = \begin{array}{c|ccc} & z_1 & z_2 & z_3 \\ \hline x_1 & 0.6 & 0.5 & 0.4 \\ x_2 & 0.7 & 0.5 & 0.3 \end{array}$

(d) $T = \begin{array}{c|ccc} & z_1 & z_2 & z_3 \\ \hline x_1 & 0.6 & 0.5 & 0.4 \\ x_2 & 0.7 & 0.7 & 0.7 \end{array}$

Ans : (c)

$$R = \begin{array}{c|cc} & y_1 & y_2 \\ \hline x_1 & 0.6 & 0.4 \\ x_2 & 0.7 & 0.3 \end{array} \; \& \; S = \begin{array}{c|ccc} & z_1 & z_2 & z_3 \\ \hline y_1 & 0.8 & 0.5 & 0.1 \\ y_2 & 0.0 & 0.6 & 0.4 \end{array}$$

To find a relation between X & Z is need to use y.
Since x_1 is related to y_1 & y_2 & y_1, y_2 are related with z_1, z_2, z_3 so relation between ×1 & z_1, z_2, z_3 can be established and so between x_2 & z_1, z_2, z_3 so for max min composition
(x_1, z_1) = max (min(0.6, 0.8) min (0.4, 0.0)) = max (0.6, 0.0) = 0.6
(x_1, z_2) = max (min (0.6, 0.5), min (0.4, 0.6)) = 0.5
(x_1, z_3) = max (min(0.6, 0.1), min (0.4, 0.4)) = 0.4
x_2z_1 = max (min(0.7, 0.8), min (0.3, 0.0)) = 0.7
x_2z_2 = max (in(0.7, 0.5), min (0.3, 0.6)) = 0.5
x_3z_3 = max (min (0.7, 0.1), min (0.3, 0.4)) = 0.3

$$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array}\begin{bmatrix} z_1 & z_3 & z_4 \\ 0.6 & 0.5 & 0.4 \\ 0.7 & 0.5 & 0.3 \end{bmatrix}$$

72. A neuron with 3 inputs has the weight vector $[0.2\ -0.1\ 0.1]^T$ and a bias $\theta = 0$. If the input vector is $X = [0.2\ 0.4\ 0.2]^T$ then the total input to the neuron is :

(a) 0.20 (b) 1.0
(c) 0.02 (d) –1.0

Ans : (c) Neural networks is a huge topic which cannot be explained in short. It is off topic anyway.
$f(x) - w^T * x + 0$
$f(x) = 0.2 * 0.2 - 0.1 * 0.4 + 0.1 * 0.2 = 0.02$

73. Which of the following neural networks uses supervised learning?
(A) Multilayer perception
(B) Self organizing feature map
(C) Hopfield network

(a) (A) only (b) (B) only
(c) (A) and (B) only (d) (A) and (C) only

Ans: (a) Multilayer perceptron network and support vector machines are examples of **supervised learning,** while self organizing feature maps and Hopfield networks are examples of **unsupervised learning.**

74. Unix command to change the case of first three lines of file "shortlist" from lower to upper

(a) $ tr '[a – z]' '[A – Z]' shortlist | head-3
(b) $ head-3 shortlist | tr '[a – z]' '[A – Z]'
(c) $ tr head-3 shortlist '[A – Z]' '[a – z]'
(d) $ tr shortlist head-3 '[a–z]' '[A – Z]'

Ans : (b) Unix command to change the case of first three lines of file "shortlist" from lower to upper are

$ head – 3 shortlist ¦ tr '[a – z]' '[A – Z]'.

75. Match the following vi commands in Unix :

List-I		List-II	
A.	**: w**	**i.**	**saves the file and quits editing mode**
B.	**: x**	**ii.**	**escapes Unix shell**
C.	**: q**	**iii.**	**saves file and remains in editing mode**
D.	**: sh**	**iv.**	**quits editing mode and no changes are saved to the file**

Codes :

	A	B	C	D
(a)	ii	iii	i	iv
(b)	iv	iii	ii	i
(c)	iii	iv	i	ii
(d)	iii	i	iv	ii

Ans : (d)
(a) w to save your file but not quit vi (iii)
(b) x <return> quit vi, writing out modified file to file named in original invocation (i)
(c) Q to quit if you haven't made any edits (iv)
(d) Sh escape the Unix shell (ii)